Continuous Assessment 2 (CA 2) - Questions

Course Code: CSE563	Course Title: APPLIED MATHEMATICS
Section(s): K3P25RB	Syllabus Covered: Unit 3 and Unit 4
Total Marks: 30 Time Allowed: 60 minutes	Maps to CO3 and CO4 - probability and statistics; graph theory

Unit III: Probability and Statistics · Unit IV: Graph Theory

Instructions: attempt all questions. Show every calculation step clearly. Given: t(0.025, 24) = 2.064, z(0.025) = 1.96 and $e^{-3} \approx 0.049787$. The marks for each question are shown in brackets.

Part A - Short Answer Questions (6 Marks - 3 questions x 2 marks)

1. A web server receives API requests following a Poisson distribution with mean $\lambda = 3$ requests per minute. Compute: (a) the probability that exactly 2 requests arrive in a given minute; and (b) the probability that at least 1 request arrives in a minute. **(2 Marks)**

2. A connected planar graph G has 8 vertices, each of degree 3 (3-regular). (a) Use the handshaking lemma to find the number of edges E. (b) Using Euler's formula ($V - E + F = 2$), find the number of faces F. **(2 Marks)**

3. Examine vertex colouring: (a) State the chromatic number $\chi(K_{3,4})$ and explain why bipartite graphs are 2-colourable. (b) State the chromatic number $\chi(C_5)$ and explain why an odd cycle cannot be 2-coloured. **(2 Marks)**

Part B - Core Numerical Problems (12 Marks - 2 questions x 6 marks)

4. In a CI pipeline, Team A submits 70 per cent of commits with a 4 per cent defect rate; Team B submits 30 per cent with a 10 per cent defect rate. **(6 Marks)**

(a) Using the law of total probability, calculate the overall probability P(D) that a random commit has a defect.

(b) If a critical defect is flagged, calculate the posterior probability P(B | D) that it came from Team B via Bayes' theorem.

(c) Compare the prior P(B) with the posterior P(B | D) and explain mathematically why the probability increases substantially.

5. Consider the weighted directed graph with vertices V = {A, B, C, D, E} and edges A→B (4), A→C (2), B→C (1), B→D (5), C→D (8), C→E (10), D→E (2). **(6 Marks)**

(a) Execute Dijkstra's single-source shortest-path algorithm from A. Present the complete iteration table showing tentative distances d[v] and predecessors, indicating which vertex becomes permanent at each step.

(b) State the shortest-path distances from A to all vertices, and the vertex sequence on the shortest path from A to E.

(c) Specify the set of directed edges forming the resulting shortest path tree.

Part C - Applied and Analytical Problems (12 Marks - 2 questions x 6 marks)

6. Statistical hypothesis testing and least-squares regression: **(6 Marks)**

(a) Benchmark query latency for $n = 25$ runs yields sample mean $\bar{x} = 142$ ms and standard deviation $s = 20$ ms. (i) Compute the standard error of the mean. (ii) Construct a 95 per cent confidence interval for the true mean μ, given $t(0.025, 24) = 2.064$. (iii) Perform a two-tailed t-test at $\alpha = 0.05$ of H_0: $\mu = 150$ ms against H_1: $\mu \neq 150$ ms. State the test statistic and your decision.

(b) Cache size x (GB) against hit rate y (per cent): (1, 50), (2, 60), (3, 70), (4, 80), (5, 90). Compute the least-squares regression line, determine R^2, and predict the hit rate at $x = 3.5$ GB.

7. Network flow optimisation and the max-flow min-cut theorem. Consider the network with source s, sink t, nodes u and v, and capacities c(s,u) = 10, c(s,v) = 8, c(u,v) = 2, c(u,t) = 7, c(v,t) = 10. **(6 Marks)**

(a) Apply the Ford-Fulkerson method. For each augmentation, state the path, the bottleneck capacity, and the updated edge flows.

(b) State the value of the maximum network flow.

(c) Find the reachable set S from s in the residual graph, define the minimum cut (S, T), compute its capacity, and verify that max-flow equals min-cut.

----- End of Question Paper -----